EXERCISE 1

An n-th Order Equation as a First-Order System and Its Uniqueness

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Reduction to a System • Existence and Uniqueness

STATEMENT

By choosing an appropriate collection of new variables $x_1, \dots, x_n$, rewrite the nth order differential equation

$$\frac{d^n x}{dt^n} = f\left(\frac{d^{n-1}x}{dt^{n-1}}, \dots, \frac{dx}{dt}, x, t\right)$$

as a set of n coupled linear first order equations. Find the conditions on the function $f(x_n, \dots, x_1, t)$ for the original differential equation to have a unique solution.

SETUP AND HYPOTHESES

Unknown: $x(t)$, n times differentiable on an interval I. Data: $f(x_n, \dots, x_1, t)$ defined on a region $\Omega \subset \mathbb{R}^{n+1}$. The initial value problem prescribes the n numbers $x(t_0), \dot{x}(t_0), \dots, x^{(n-1)}(t_0)$. The tool for uniqueness is the existence and uniqueness theorem for first-order systems $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}, t)$ (Robinson Thm 25.2, Topic 4 Sec. 1).

FORMULATION: THE NEW VARIABLES

Introduce one variable for the function and for each derivative below the highest order (Robinson Sec. 25.1):

$$x_1 = x, \quad x_2 = \frac{dx}{dt}, \quad x_3 = \frac{d^2x}{dt^2}, \quad \dots, \quad x_n = \frac{d^{n-1}x}{dt^{n-1}}, \qquad \text{i.e. } x_k = x^{(k-1)} \ (k = 1, \dots, n)$$

METHOD OF THE CLASS NOTES [NOTES, P. 9]

The important point of the notes: an ODE of order $n \geq 2$ is a system of n first-order ODEs; for $\ddot{x} = f(\dot{x}, x, t)$ they set $x_1 = x$, $x_2 = \dot{x}$, so that $\dot{x}_1 = x_2$ and $\dot{x}_2 = f(x_2, x_1, t)$. The variables above are the same choice for order n. Uniqueness then follows from the condition the notes state for systems: $\mathbf{f}$ and its Jacobian $D\mathbf{f}$ continuous near $(\mathbf{x}_0, t_0)$. Only the last component f is not trivial, so the condition falls on f and its partial derivatives $\partial f / \partial x_k$.

DERIVATION

The system

By construction $\dot{x}_k = x^{(k)} = x_{k+1}$ for $k = 1, \dots, n-1$, and $\dot{x}_n = x^{(n)} = f(x_n, \dots, x_1, t)$ by the original equation:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ &\vdots \\ \dot{x}_{n-1} &= x_n \\ \dot{x}_n &= f(x_n, x_{n-1}, \dots, x_1, t) \end{aligned} \qquad \Longleftrightarrow \qquad \dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}, t) = \begin{pmatrix} x_2 \\ x_3 \\ \vdots \\ x_n \\ f(x_n, \dots, x_1, t) \end{pmatrix}$$

The first $n-1$ equations are linear in the unknowns (they say $\dot{x}_k = x_{k+1}$); the last one is linear exactly when f is a linear function of $x_1, \dots, x_n$ (as for a linear ODE, where the whole system is linear, $\dot{\mathbf{x}} = A(t)\mathbf{x} + \mathbf{b}(t)$). The first-order system form is valid for any f.

Equivalence

(a) If x solves the nth-order equation, then $\mathbf{x} = (x, \dot{x}, \dots, x^{(n-1)})^T$ solves the system. (b) Conversely, if $(x_1, \dots, x_n)$ solves the system, then $\dot{x}_1 = x_2, \ddot{x}_1 = \dot{x}_2 = x_3, \dots, x_1^{(k)} = x_{k+1}$ for $k \leq n-1$, and $x_1^{(n)} = \dot{x}_n = f(x_n, \dots, x_1, t) = f(x_1^{(n-1)}, \dots, \dot{x}_1, x_1, t)$: the function $x = x_1$ solves the original equation. The initial condition $\mathbf{x}(t_0) = (x_1^0, \dots, x_n^0)$ is exactly $x(t_0) = x_1^0, \dot{x}(t_0) = x_2^0, \dots, x^{(n-1)}(t_0) = x_n^0$.

Conditions for a unique solution

Theorem 25.2 (Robinson p. 250) requires $\mathbf{F}$ and the matrix of partial derivatives $D\mathbf{F}$ to be continuous on an open set containing the initial point. Compute it row by row: for $F_k = x_{k+1}$ ($k < n$) the partial derivatives are 0 or 1, and for $F_n = f$ they are $\partial f / \partial x_j$:

$$D\mathbf{F} = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ \dfrac{\partial f}{\partial x_1} & \dfrac{\partial f}{\partial x_2} & \dfrac{\partial f}{\partial x_3} & \cdots & \dfrac{\partial f}{\partial x_n} \end{pmatrix}$$

The first $n-1$ rows are constant, hence continuous. Therefore all the conditions reduce to the last row:

CONDITION FOR UNIQUENESS

$f(x_n, \dots, x_1, t)$ and its partial derivatives $\partial f/\partial x_j$, $j = 1, \dots, n$, must be continuous in an open region containing the initial point $(x(t_0), \dots, x^{(n-1)}(t_0), t_0)$. Then the IVP has exactly one solution on some interval around t_0 (local existence and uniqueness), and two solutions with the same $x(t_0), \dots, x^{(n-1)}(t_0)$ coincide wherever both exist.

INTERPRETATION AND CHECKS

- For $n = 2$ the system is $\dot{x}_1 = x_2, \dot{x}_2 = f(x_2, x_1, t)$ and the condition is Theorem 11.1 of Topic 3 (continuity of $f, \partial f/\partial x_1, \partial f/\partial x_2$); for $n = 1$ it is Picard's theorem.
- No condition on $\partial f/\partial t$ is needed: the theorem asks continuity in t but no differentiability in t.
- Dependence on the n derivatives: continuity of the partial derivatives $\partial f/\partial x_j$ gives a local Lipschitz bound in $(x_1, \dots, x_n)$, which is what excludes branching solutions (compare Exercise 2.14 of Chapter 2, where the Lipschitz condition fails).
- For a linear equation $x^{(n)} + a_{n-1}(t)x^{(n-1)} + \cdots + a_0(t)x = g(t)$, $f = g - \sum a_k x_{k+1}$ and $\partial f/\partial x_{k+1} = -a_k(t)$: the conditions reduce to continuity of the a_k and g. For linear equations the solution in fact exists on the whole interval of continuity of the coefficients (a standard result for linear systems, not in Robinson Thm 25.2, which is local; see the continuation argument in Exercise 4).
- Check with the mass-spring-damper (Topic 4 Sec. 1.1): $n = 2$, $f = -\lambda x_2 - \omega^2 x_1$ has constant partial derivatives $-\omega^2$ and $-\lambda$: unique solutions for all data.